

9 HEAT STRUCTURES

These are used in NEW and RESTART type problems and are required only if heat structures are described. The heat structure card numbers are of the form 1CCCGXNN where the fields are defined as follows:

CCC is a heat structure number. The heat structure numbers need not be consecutive. We suggest, but the system does not require, that if heat structures and hydrodynamic volumes are related, they be given the same number.

G is a geometry number. The combination CCCG is a heat structure geometry combination referenced in the heat structure input data. The G digit is provided to differentiate between different types of heat structures (such as fuel pins and core barrel) that might be associated with the same hydrodynamic volume.

X is the card type.

NN is the card number within a card type.

9.1 General Heat Structure Data, 1CCCG000

This card is required for heat structures. Use 8 words for new data input or 1 word for deleting a heat structure.

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| W1(I) | Number of axial heat structures with this geometry, nh. This number must be > 0 and < 100. |
| W2(I) | Number of radial mesh points for this geometry, np. This number must be < 100. Enter > 1 if no reflood is specified, and > 2 if reflood or metal water reaction is specified. |
| W3(I) | Geometry type. Enter 1 for rectangular, 2 for cylindrical, and 3 for spherical. Spherical geometry is not allowed if reflood is specified. Cylindrical geometry must be specified when the gap conductance model is used. |
| W4(I) | Steady-state initialization flag. Use 0 if the desired initial condition temperatures are entered on input cards 1CCCG401-G499; use 1 if the steady-state initial condition temperatures are to be calculated by the code. If option 1 is chosen, the user is still required to enter temperatures on cards 1CCCG402-G499. In this case, the temperatures are used as starting points for the steady-state solutions. The user should therefore enter temperatures either below or above the minimum film boiling point to assure the respective pre-DNB or post-DNB steady-state condition is calculated. This is because the boiling curve is multi-valued. |

- W5(R) Left boundary coordinate (m, ft).
- W6(I) Reflood condition flag. This quantity is optional if no reflood calculation is to be performed. This quantity may be 0, 1, 2, or a trip number.
- If 0, no reflood calculation is to be performed.
- If nonzero, all the heat structures in this heat structure/geometry are assumed to form a two-dimensional representation of a fuel pin. The radial mesh is defined on card 1CCCG1NN. Each heat structure represents an axial level of the fuel pin, with the first heat structure being the bottom level. Each heat structure should be connected to a hydrodynamic volume representing the same axial section of the coolant channel. The length of the axial mesh in the fuel pin is given by the height of the connected hydrodynamic volume. If the heat structure is fuel pins or heat exchanger tubes, the length factor (Word 5 on cards 1CCCG501-G509) is the product of the hydrodynamic volume length and the number of pins or tubes. The heat structures represent the temperatures at the midpoint of the axial mesh. Once the reflood calculation is initiated, additional mesh lines are introduced at each end of the fuel pin and between the heat structures. Once the reflood calculation is initiated, it remains activated, and the two-dimensional heat conduction calculation uses a minimum of $2 \bullet nh + 1$ axial mesh nodes. Additional mesh lines are introduced and later eliminated as needed to follow the quench front.
- If 1 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa, and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.9 (i.e., nearly empty).
- If 2 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.1 (i.e., dryout begins).
- If a trip number is entered, the reflood calculation is initiated when the trip is set true. When using the expanded trip number format, 1 and 2 are possible trip numbers. A 1 or 2 entered in this word is not treated as a trip number.
- W7(I) Boundary volume indicator. Enter 0 or 1 to indicate that reflood heat transfer applies to the left or right boundary, respectively.
- W8(I) Maximum number of axial intervals. Enter 2, 4, 8, 16, 32, 64, or 128 to indicate the maximum number of axial subdivisions a heat structure can have. Storage is allocated for the number indicated, even though a transient may not require that level of subdivision.